

EXPERIMENT 5

PLAYFAIR CIPHER

AIM

To implement the Playfair Cipher substitution technique using Python.

ALGORITHM

1. Read the plaintext and keyword.
2. Construct the 5×5 key matrix.
3. Split the plaintext into pairs.
4. Apply Playfair encryption rules.
5. Display the encrypted text.

PYTHON CODE

```
def generate_key_table(key):  
    key = key.upper().replace("J", "I")  
    table = []  
    for c in key:  
        if c not in table and c.isalpha():  
            table.append(c)  
    for c in "ABCDEFGHIKLMNOPQRSTUVWXYZ":  
        if c not in table:  
            table.append(c)  
    return [table[i:i+5] for i in range(0, 25, 5)]  
  
def find(table, ch):  
    for i in range(5):  
        for j in range(5):  
            if table[i][j] == ch:  
                return i, j  
  
def encrypt(text, table):
```

```
text = text.upper().replace("J", "I")
text = "".join(c for c in text if c.isalpha())
```

```
pairs = []
i = 0
while i < len(text):
    a = text[i]
    if i + 1 < len(text):
        b = text[i + 1]
        if a == b:
            pairs.append((a, "X"))
            i += 1
        else:
            pairs.append((a, b))
            i += 2
    else:
        pairs.append((a, "X"))
        i += 1
```

```
cipher = ""
```

```
for a, b in pairs:
    r1, c1 = find(table, a)
    r2, c2 = find(table, b)

    if r1 == r2:
        cipher += table[r1][(c1 + 1) % 5]
        cipher += table[r2][(c2 + 1) % 5]
    elif c1 == c2:
```

```

        cipher += table[(r1 + 1) % 5][c1]

        cipher += table[(r2 + 1) % 5][c2]

    else:

        cipher += table[r1][c2]

        cipher += table[r2][c1]

    return cipher

key = input("Enter Key: ")
plain = input("Enter Plain Text: ")

table = generate_key_table(key)

print("Encrypted Text:", encrypt(plain, table))

```

OUTPUT

```

IDLE Shell 3.14.2
File Edit Shell Debug Options Window Help
Python 3.14.2 (tags/v3.14.2:df79316, Dec 5 2025, 17:18:21) [MSC v.1944 64 bit (AMD64)] on win32
Enter "help" below or click "Help" above for more information.
>>>
=== RESTART: C:/Users/preet/OneDrive/Desktop/csa5051/lab experiments/LAB5.PY ===
Enter Key: KEYWORD
Enter Plain Text: MONARCH
Encrypted Text: SKPDDRIV
>>>
=== RESTART: C:/Users/preet/OneDrive/Desktop/csa5051/lab experiments/LAB5.PY ===
Enter Key: LIBERTY
Enter Plain Text: EDUCATION
Encrypted Text: RCXTCYYVQU
>>>

```

RESULT

Thus the Playfair Cipher substitution technique was implemented successfully using Python.